

Data Directory

This directory is intended for storing blood glucose datasets used in benchmark experiments.

Directory Structure

```
data/
├── raw/                # Original, unprocessed datasets
│   ├── ohiot1dm/      # Ohio T1DM Dataset
│   └── custom/         # Custom datasets
├── processed/          # Preprocessed and cleaned datasets
└── metadata/           # Dataset descriptions and schemas
```

Supported Datasets

OhioT1DM Dataset

- **Description:** Type 1 diabetes dataset with continuous glucose monitoring
- **Source:** Ohio University
- **Format:** CSV files with CGM, insulin, meal, and activity data
- **Subjects:** Multiple patients with different data availability

Custom Datasets

- **Description:** User-provided datasets
- **Format:** CSV with standardized column names
- **Required columns:** timestamp, subject_id, glucose
- **Optional columns:** insulin, carbs, activity, etc.

Data Format Requirements

All datasets should be converted to a standardized format with the following columns:

Required Columns

- **timestamp:** ISO format datetime (YYYY-MM-DD HH:MM:SS)
- **subject_id:** Unique identifier for each subject
- **glucose:** Blood glucose value in mg/dL

Optional Columns

- **insulin:** Insulin dose in units
- **carbs:** Carbohydrate intake in grams
- **activity:** Activity level or type
- **sleep:** Sleep status or quality

- **stress**: Stress level
- **heart_rate**: Heart rate in BPM

Usage

Data loading is handled automatically by the benchmark framework:

```
from benchmark.data.loaders import load_dataset

# Load dataset
data = load_dataset('ohiot1dm', subjects=['559', '563'])

# Data is returned in standardized format
print(data.columns) # ['timestamp', 'subject_id', 'glucose', ...]
```

Adding New Datasets

1. Place raw data in appropriate subdirectory
2. Implement loader in **benchmark/data/loaders.py**
3. Add validation rules in **benchmark/data/validators.py**
4. Update configuration schema
5. Add documentation here

Notes

- This directory is excluded from version control via **.gitignore**
- Large datasets should be stored externally and downloaded as needed
- Always include proper licensing and attribution for datasets